

Cognitive Integrity Framework: Trust Calculus and Detection Bounds

Cognitive Integrity Framework: Trust Calculus and Detection Bounds I

This section presents the formal foundations of the Cognitive Integrity Framework (CIF). We define the system model (`sec:system-model`), cognitive state representation (`sec:cognitive-state`), integrity properties (`sec:integrity-properties`), trust calculus (`sec:trust-calculus`), and information-theoretic detection bounds (`sec:detection-bounds`).

System Model I

Definition (Multiagent Operator)

A *multiagent operator* is a tuple:

$$\mathcal{O} = \langle \mathcal{A}, \mathcal{C}, \mathcal{S}, \mathcal{P}, \Gamma \rangle \quad (1)$$

where components are defined in `tab:operator-components`.

Table 1: Components of the multiagent operator $\mathcal{O}$.

Component	Symbol	Description
Agents	$\mathcal{A} = \{a_1, \dots, a_n\}$	Finite set of n agents
Communication	$\mathcal{C} : \mathcal{A} \times \mathcal{A} \rightarrow \{0, 1\}$	Adjacency matrix encoding permitted channels
Shared State	$\mathcal{S}$	Observable global state
Permissions	$\mathcal{P} : \mathcal{A} \times \text{Actions} \rightarrow \{0, 1\}$	Action authorization mapping
Protocol	Γ	Coordination and communication rules

Cognitive State I

Intuitively, an agent's cognitive state captures everything it believes, wants, intends, and remembers at a given moment. This formal representation enables precise reasoning about how attacks manipulate agent reasoning.

Definition (Agent Cognitive State)

Each agent $a_i \in \mathcal{A}$ maintains cognitive state:

$$\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{I}_i, \mathcal{H}_i \rangle \quad (2)$$

with components defined in `tab:cognitive-components`.

Table 2: Cognitive state components for agent a_i .

Component	Formal Type	Semantics
Beliefs	$\mathcal{B}_i : \Phi \rightarrow [0, 1]$	Probability distribution over propositions
Goals	$\mathcal{G}_i = \{(g_k, p_k)\}$	Prioritized objectives where $\sum_k p_k = 1$
Intentions	$\mathcal{I}_i = [(a_1, t_1), \dots]$	Committed action sequence with timing
History	$\mathcal{H}_i = [(e_1, t_1), \dots]$	Interaction trace (events, timestamps)

Cognitive State II

Definition (System State)

The global system state at time t is:

$$S^t = (\sigma_1^t, \dots, \sigma_n^t, \mathcal{S}^t, \mathcal{T}^t) \quad (3)$$

where $\mathcal{T}^t$ denotes the trust matrix at time t .

Cognitive State III

State Transition Semantics

The following transition rules formalize how agent states evolve. Each rule has the form “if preconditions hold (above the line), then this transition occurs (below the line).” Readers may skim the mathematical details on first reading, returning for precision when needed.

Definition (Transition Relation)

State transitions follow the relation $S^t \xrightarrow{\alpha} S^{t+1}$ where $\alpha \in \{\text{RECEIVE}, \text{UPDATE}, \text{ACT}, \text{COMMUNICATE}\}$.

The transition rules are defined as follows:

Rule T-Receive (Message Reception):

$$\frac{m \in \text{channel}(a_j, a_i) \quad \mathcal{F}(m) = \text{ACCEPT}}{(\sigma_i, \text{inbox}_i) \xrightarrow{\text{RECEIVE}} (\sigma_i, \text{inbox}_i \cup \{m\})} \quad (4)$$

Rule T-Reject (Message Rejection):

$$m \in \text{channel}(a_j, a_i) \quad \mathcal{F}(m) \in \{\text{REJECT}, \text{QUARANTINE}\} \quad (5)$$

Integrity Properties I

We define four core integrity properties that CIF aims to preserve.

[Belief Consistency]

$$\text{Consistent}(\mathcal{B}_i) \iff \nexists \phi, \psi : \mathcal{B}_i(\phi) > \tau \wedge \mathcal{B}_i(\psi) > \tau \wedge (\phi \wedge \psi \vdash \perp) \quad (9)$$

No high-confidence beliefs contradict each other.

[Goal Alignment]

$$\text{Aligned}(\mathcal{G} * i) \iff \mathcal{G} * i \subseteq \mathcal{G} * \text{principal} \cup \text{Delegate}(\mathcal{G} * \text{principal}) \quad (10)$$

All goals derive from the principal or valid delegation chains.

[Provenance Verifiability]

$$\text{Verifiable}(\mathcal{B}_i) \iff \forall \phi : \mathcal{B}_i(\phi) > \tau \Rightarrow \exists \pi(\phi) : V(\pi(\phi)) = 1 \quad (11)$$

Every accepted belief has a verifiable provenance chain π .

Integrity Properties II

[Action Authorization]

$$\text{Auth}(a_i, \text{act}) \iff \mathcal{P}(a_i, \text{act}) = 1 \vee \exists a_j : \text{Delegate}(a_j, a_i, \text{act}) \quad (12)$$

Actions require direct permission or valid delegation.

Trust Calculus I

Trust Calculus II

Motivation: Why Bounded Trust Matters

Before presenting the formal trust calculus, we motivate its design through concrete scenarios that illustrate why naive trust models fail in multiagent systems.

The Trust Laundering Problem. Consider an adversary with low direct trust who seeks to influence a high-value agent. In a naive trust model, the adversary could:

1. Establish contact with a moderately trusted intermediary agent
2. Provide accurate information over time to build trust with the intermediary
3. Use the intermediary to relay adversarial content to the target
4. The target accepts the content because it comes from a “trusted” source

This is *trust laundering*—converting low-trust origin into high-trust delivery through intermediaries. Without bounded delegation, the adversarial content arrives at the target with the intermediary’s trust score, not the adversary’s.

The Trust Amplification Problem. In peer-to-peer multiagent

Information-Theoretic Detection Bounds I

Having established the trust calculus (how agents reason about each other) and belief update semantics (how agents incorporate information), we now turn to fundamental limits on attack detection. This section establishes fundamental limits on what any detection system can achieve. Like Shannon's channel capacity in communications, these bounds are not limitations of specific mechanisms but mathematical constraints on what is possible.

Definition (Attack Information Channel)

An attack models a communication channel from adversary to target:

$$\text{Channel} : \mathcal{A} * \text{adv} \rightarrow \sigma * \text{target} \quad (30)$$

Information-Theoretic Detection Bounds II

Theorem (Minimum Attack Entropy)

For attack $\mathcal{A}$ to succeed with probability p :

$$H(\mathcal{A}) \geq -\log_2(1 - p) + H(\sigma_{\text{target}}|\mathcal{A}) \quad (31)$$

Proof.

By the data processing inequality. The attack must contain sufficient information to change the target state. □

Corollary

Attacks with low entropy (simple patterns) are more detectable.

Definition (Detector Information Gain)

For detector D observing system state S :

$$I(D; \mathcal{A}) = H(\mathcal{A}) - H(\mathcal{A}|D(S)) \quad (32)$$

Information-Theoretic Detection Bounds III

Theorem (Fundamental Detection Limit)

No detector achieves detection rate r if:

$$r > \frac{I(D; \mathcal{A})}{H(\mathcal{A})} \quad (33)$$

The following theorem captures the fundamental tradeoff facing attackers: high-impact attacks are easier to detect, while stealthy attacks have limited effect. This is not a limitation of our defenses—it is a mathematical constraint that any attack must satisfy.

Information-Theoretic Detection Bounds IV

Theorem (Stealth-Impact Tradeoff)

For attack with impact $\mathcal{I}$ and stealth $\mathcal{S}$ (inverse detectability):

$$\mathcal{I} \cdot \mathcal{S} \leq C_{\text{channel}} \quad (34)$$

where C_{channel} is the attack channel capacity.

[Geometric Sharpening of the Stealth-Impact Bound] Theorem ?? can be sharpened using the information geometry of the belief manifold (Part 2, Supplement S10). When belief space is endowed with the Fisher-Rao metric $G_{ij}(p) = \delta_{ij}/p_i$, the Fisher-Rao geodesic distance

$$d_{\text{FR}}(p, q) = 2 \arccos \left(\sum_i \sqrt{p_i q_i} \right)$$

measures the geometric “distance” of an attack. Impact $\mathcal{I}$ is proportional to $d_{\text{FR}}(p_0, p_{\text{attacked}})$, while stealth $\mathcal{S}$ is inversely

Information-Theoretic Detection Bounds V

proportional to the same quantity. The curvature of the probability simplex Δ^{n-1} (constant positive curvature $\kappa = n(n-1)/4$) implies that the product $\mathcal{J} \cdot \mathcal{S}$ is bounded by the diameter of the manifold:

$$\mathcal{J} \cdot \mathcal{S} \leq \frac{\pi}{2},$$

since $d_{\text{FR}} \leq \pi$ (Hellinger antipodal distributions). This recovers $C_{\text{channel}} = \pi/2$ as the tight bound when impact is measured in natural (Fisher-Rao) units, providing a geometric proof of Theorem ?? with the explicit constant. Furthermore, the connection to KL divergence— $D_{\text{KL}}(p \parallel p') \approx \frac{1}{2}(\Delta p)^\top G(p)(\Delta p)$ for small perturbations—gives geometric justification for the drift detection threshold $\theta_{\text{drift}} = 0.3$: this corresponds to a Fisher-Rao step of approximately 0.28 radians, or roughly 9% of the maximum possible attack distance (Part 2, Supplement S10).

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Table 5: Information-theoretic bounds by attack type.

Attack Type	Min Entropy $H(\mathcal{A})$	Detection Lower Bound
Belief Injection	$\log_2 \Phi $	$\frac{\log_2 \Phi }{H(\mathcal{B})}$
Goal Hijacking	$H(\mathcal{G}_{\text{target}})$	$\frac{H(\mathcal{G}_{\text{target}})}{H(\mathcal{G})}$
Trust Manipulation	$\log_2 n$	$\frac{\log_2 n}{H(\mathcal{T})}$
Progressive Drift	$T \cdot \log_2(1/\delta_{\text{step}})$	$\frac{T}{w} \cdot \delta_{\text{step}}$

Theorem (Progressive Attack Detection Bound)

For progressive drift attack with step size δ over T steps:

$$P(\text{detect within } w \text{ steps}) \leq 1 - \left(1 - \frac{\delta}{\theta_{\text{step}}}\right)^w \quad (35)$$

Corollary

Smaller step sizes exponentially reduce detection probability but require more time for impact.

Information-Theoretic Detection Bounds VII

Definition (Defense Information Budget)

Total monitoring capacity: $B_{\text{total}} = \sum_{d \in \mathcal{D}} B_d$.

Theorem (Optimal Budget Allocation)

Given attack distribution $P(\mathcal{A}_k)$ and detector sensitivities $\eta_d(\mathcal{A}_k)$:

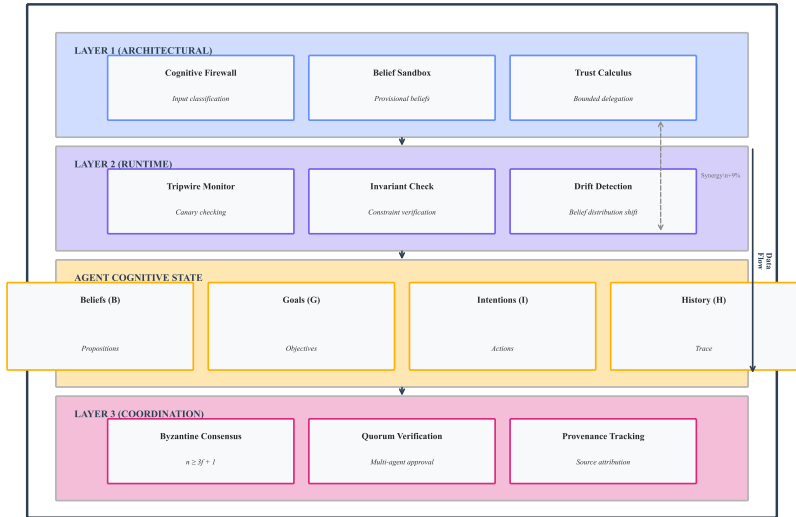
$$B_d^* = \frac{\sum_k P(\mathcal{A}_k) \cdot \eta_d(\mathcal{A}_k)}{\sum_{d'} \sum_k P(\mathcal{A}_k) \cdot \eta_{d'}(\mathcal{A}_k)} \cdot B_{\text{total}} \quad (36)$$

Proof.

Lagrangian optimization maximizing expected detection subject to budget constraint. □

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COGNITIVE INTEGRITY FRAMEWORK (CIF)



Information-Theoretic Detection Bounds IX

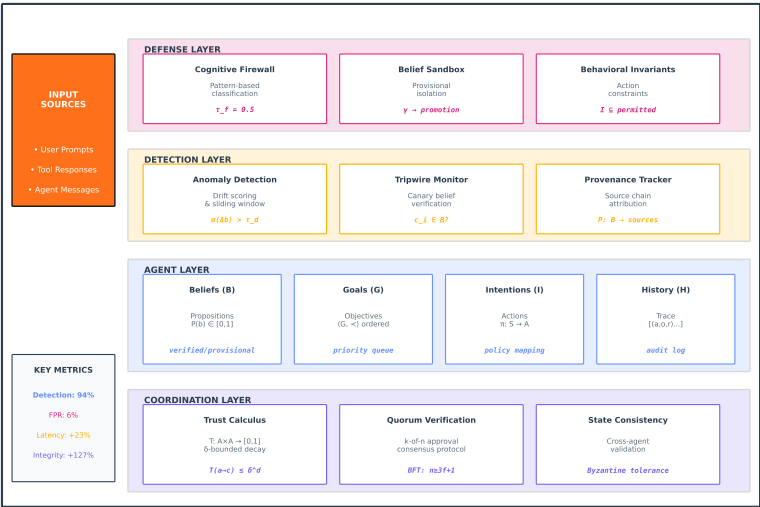
Figure 4: CIF Architecture Overview: Three-layer defense architecture—**Layer 1** (Architectural): Cognitive Firewall at entry, Belief Sandbox for unverified data, Trust Calculus for delegation; **Layer 2** (Runtime): Tripwires for belief monitoring, Invariant verification, Drift detection; **Layer 3** (Coordination): Byzantine consensus, Quorum verification, Provenance tracking.

`fig:cif-architecture` presents the layered CIF architecture with architectural defenses (left), runtime defenses (center), and coordination mechanism (right). `fig:cif-comprehensive` expands this to show data flow, attack interception points, and defense composition.

Information-Theoretic Detection Bounds X

COGNITIVE INTEGRITY FRAMEWORK (CIF)

Layered Defense Architecture for Multiagent AI Systems



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Figure 5: Comprehensive CIF Architecture: Extended architecture showing data flow from user input through all defense layers to agent output. Attack interception points labeled Ω_1 – Ω_5 indicate where each adversary class is detected. Defense composition follows multiplicative detection rate improvement (cor:layered-defense).

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`fig:cif-comprehensive` provides a detailed view of the complete CIF architecture, including all component formulas and their interactions. The defense layer implements the cognitive firewall with threshold $\tau_f = 0.5$, the belief sandbox with promotion function γ , and behavioral invariants constraining intentions $\mathcal{I} \subseteq \text{permitted}$. The detection layer specifies anomaly scoring $\sigma(\Delta b) > \tau_d$, tripwire verification $c_i \in \mathcal{B}?$, and provenance tracking $P : \mathcal{B} \rightarrow \text{sources}$. The coordination layer encodes the trust calculus $\mathcal{T} : \mathcal{A} \times \mathcal{A} \rightarrow [0, 1]$ with δ -bounded decay, k-of-n quorum protocols, and Byzantine fault tolerance ($n \geq 3f + 1$). For empirical validation of detection rates and performance overhead, see Part 2 of this series.

CIF-AD Integration: Action-Delegation Coupling I

The Cognitive Integrity Framework provides static structural guarantees (trust bounds, belief consistency). But multiagent operators are dynamic: agents act, delegate, and observe outcomes. This section formalizes how CIF integrates with the Action-Delegation (AD) cycle to provide continuous runtime security guarantees.

CIF-AD Integration: Action-Delegation Coupling II

The Action-Delegation Cycle

Definition (Action-Delegation Cycle)

The Action-Delegation (AD) cycle for agent a_i at time t is:

$$\text{AD}_i^t : \sigma_i^t \xrightarrow{\text{plan}} \mathcal{J}_i^t \xrightarrow{\text{delegate}} \langle a_j, \text{task}_j \rangle \xrightarrow{\text{execute}} \mathcal{H}_i^{t+1} \xrightarrow{\text{update}} \sigma_i^{t+1} \quad (37)$$

comprising: cognitive state assessment, intention formation, delegation decision, execution observation, and state update.

Each phase of the AD cycle introduces distinct attack surfaces that CIF must protect:

Table 6: AD cycle phases and corresponding CIF defenses.

AD Phase	Attack Surface	CIF Mechanism	Formal Guarantee
Plan	Goal injection	Invariant enforcement	$\mathcal{G}_i \subseteq \mathcal{G}_{\text{principal}}$
Delegate	Trust exploitation	Trust calculus	$\mathcal{J}^{\text{del}} \leq \delta^d$
Execute	Result tampering	Provenance tracking	$V(\pi(\phi)) = 1$
Observe	Evidence fabrication	Belief sandbox	$\mathcal{B}_{\text{prov}}$ isolation
Update	Drift induction	KL drift detection	$D_{\text{KL}} < \theta_{\text{drift}}$

CIF-AD Integration: Action-Delegation Coupling III

CIF-AD Coupling Matrix

Definition (CIF-AD Coupling Matrix)

The coupling matrix $\mathbf{M}_{AD} \in \mathbb{R}^{5 \times 5}$ quantifies the degree to which each CIF defense addresses each AD phase:

$$[\mathbf{M}_{AD}]_{ij} = \text{Coverage}(\text{Defense}_i, \text{Phase}_j) \in [0, 1] \quad (38)$$

Table 7: CIF-AD coupling matrix: coverage of each defense mechanism across AD cycle phases.

Defense	Plan	Delegate	Execute	Observe	Update
Cognitive Firewall	0.80	0.40	0.20	0.90	0.30
Belief Sandbox	0.50	0.30	0.30	0.85	0.70
Trust Calculus	0.30	0.95	0.20	0.10	0.20
Tripwires + Invariants	0.90	0.20	0.80	0.40	0.60
Byzantine Consensus	0.20	0.70	0.90	0.30	0.50

Theorem (CIF-AD Full Coverage)

The CIF defense stack provides non-zero coverage for every AD cycle phase:

$$\forall j \in \{1, \dots, 5\} : \max[\mathbf{M}_{AD}]_{ij} > 0 \quad (39)$$

CIF-OODA Integration: Temporal Security Across Decision Cycles I

The OODA loop (Observe-Orient-Decide-Act) models how agents process information and take action over time. Cognitive attacks often target specific OODA phases to corrupt decision-making while minimizing detectable footprint. This section formalizes the OODA-phase attack model and CIF's phase-specific defenses.

CIF-OODA Integration: Temporal Security Across Decision Cycles II

OODA Loop Formalization

Definition (OODA State Machine)

Agent a_i 's OODA loop is a state machine:

$$\text{OODA}_i = (Q, q_0, \Sigma, \delta_{\text{OODA}}) \quad (40)$$

where $Q = \{\text{Observe, Orient, Decide, Act}\}$ are states, $q_0 = \text{Observe}$ is the initial state, $\Sigma = \mathcal{M} \cup \text{Events}$ is the input alphabet, and $\delta_{\text{OODA}} : Q \times \Sigma \rightarrow Q$ is the transition function.

The OODA loop maps to cognitive state components:

$$\text{Observe} \leftrightarrow \mathcal{H}_i \text{ (history accumulation)} \quad (41)$$

$$\text{Orient} \leftrightarrow \mathcal{B}_i \text{ (belief update)} \quad (42)$$

$$\text{Decide} \leftrightarrow \mathcal{G}_i, \mathcal{J}_i \text{ (goal selection, intention formation)} \quad (43)$$

$$\text{Act} \leftrightarrow \text{Actions}(a_i) \text{ (execution)} \quad (44)$$